MoveIt Quick Start

DR-xcobo 2.0 | v1.0 | ROS 1 / MoveIt 1 | English

Environment: Ubuntu 20.04 LTS Desktop with ROS Noetic Desktop installed. Use Bash. This guide covers motion planning and trajectory preview in RViz.

Download the original package ZIP into **~/Downloads/** and keep its filename. Only the archive and outer folder are renamed; all enclosed files are unchanged.

1. Install dependencies (once)

```
source /opt/ros/noetic/setup.bash
sudo apt update
sudo apt install unzip ros-noetic-moveit ros-noetic-rviz \
  ros-noetic-joint-state-publisher ros-noetic-robot-state-publisher
```

These commands assume ROS is installed. If the first line cannot find the file, install ROS Noetic first.

2. Extract the package (once)

```
mkdir -p ~/dr_xcobo_moveit
cd ~/dr_xcobo_moveit
unzip ~/Downloads/dr-xcobo-2.0-moveit-v1.0.zip
```

Keep both inner directories: **Dr_arm_6_desk** and **Dr_arm_6_desk_moveit**. Do not rename them or merge them with another copy.

3. Create a separate preview launcher (once)

Copy the complete block. It creates **preview.launch** outside the original packages, without editing them.

```
cat > ~/dr_xcobo_moveit/preview.launch <<'EOF'
<launch>
  <arg name="cfg" default="$(find Dr_arm_6_desk_moveit)" />
  <include file="$(arg cfg)/launch/move_group.launch">
    <arg name="allow_trajectory_execution" value="false" />
  </include>
  <node name="joint_states" pkg="joint_state_publisher"
        type="joint_state_publisher" />
  <node name="robot_state" pkg="robot_state_publisher"
        type="robot_state_publisher" />
  <include file="$(arg cfg)/launch/moveit_rviz.launch">
    <arg name="rviz_config"
         value="$(arg cfg)/launch/moveit.rviz" />
  </include>
</launch>
EOF
```

The original demo.launch uses a machine-specific path and starts a hardware bridge. Use this separate launcher, which disables execution. Continue on the next page.

Launch & Use

DR-xcobo 2.0 | RViz / OMPL | English

4. Each new terminal: load the environment & launch

```
source /opt/ros/noetic/setup.bash
export ROS_PACKAGE_PATH="$HOME/dr_xcobo_moveit/\
dr-xcobo-2.0-moveit-v1.0:$ROS_PACKAGE_PATH"
roslaunch ~/dr_xcobo_moveit/preview.launch
```

source loads ROS; **export** locates the two original packages; **roslaunch** starts planning and RViz. The backslash continues the line: copy the full block. A separate roscore is unnecessary.

This preview uses installed ROS nodes and package configuration files, with no custom compilation targets. It needs neither catkin_make nor colcon build; other ROS packages may need a build.

5. Plan a trajectory in RViz

1. Check settings: set Global Options → Fixed Frame to **base_link**, and MotionPlanning → Robot Description to **robot_description**.

2. Select the group: choose **arm** as the Planning Group. The saved configuration may show **left_arm**; change it manually. If the plugin is missing, use Add → MotionPlanning.

3. Set start and goal: select **current** as the Start State. Enable Query Goal State and drag the goal using Interact, or adjust target joints in the Joints tab. Both states must be collision-free.

4. Click Plan: view the animation under Planned Path. For step-by-step playback, open Panels → MotionPlanning - Slider, then press Play.

5. Stop: press **Ctrl+C** in the terminal. This launcher supports planning and preview only. Do not use Execute or Plan & Execute; it does not launch Gazebo or a hardware driver.

Why load the environment in each terminal?

source and export affect only the current terminal. Repeat them in a new terminal, not repeatedly in the same one. Daily use does not require reinstalling, extracting, or recreating the launcher.

Optional: add the source line and complete export command above to **~/.bashrc** (omit roslaunch), then open a new terminal. Configure only one ROS environment and avoid duplicate entries.

Troubleshooting

Package not found: check the extracted folders, exact case, and exported path; avoid another copy taking precedence.
Missing model: retain all mesh files and use base_link as the Fixed Frame.
Planning fails: select arm and a reachable, collision-free target, then click Plan.

Checked against the package files and official documentation; not run in a ROS environment. Ubuntu 20.04 / Noetic is the course compatibility target. ROS Noetic and MoveIt 1 are end-of-life.

Course resources | MoveIt 1 installation | Official RViz tutorial

MoveIt Quick Start

DR-xcobo 2.0 | v1.0 | ROS 1 / MoveIt 1 | 中文版

环境：Ubuntu 20.04 LTS 桌面版，已安装 ROS Noetic Desktop；使用 Bash 终端。本教程用于 RViz 运动规划和轨迹预览。

下载原始功能包 ZIP，保留文件名并放在 ~/Downloads/。包内文件原样保留；只规范了 ZIP 和最外层目录名称。

1. 安装依赖（首次使用）

```
source /opt/ros/noetic/setup.bash
sudo apt update
sudo apt install unzip ros-noetic-moveit ros-noetic-rviz \
  ros-noetic-joint-state-publisher ros-noetic-robot-state-publisher
```

以上命令不安装完整 ROS；若第一行提示文件不存在，请先安装 ROS Noetic。

2. 解压（首次使用）

```
mkdir -p ~/dr_xcobo_moveit
cd ~/dr_xcobo_moveit
unzip ~/Downloads/dr-xcobo-2.0-moveit-v1.0.zip
```

解压后保留两个内部目录：Dr_arm_6_desk 和 Dr_arm_6_desk_moveit。不要改名或与已有的同名包合并。

3. 创建独立预览入口（首次使用）

完整复制以下代码块。它在原包外创建 preview.launch，不修改原包。

```
cat > ~/dr_xcobo_moveit/preview.launch <<'EOF'
<launch>
  <arg name="cfg" default="$(find Dr_arm_6_desk_moveit)" />
  <include file="$(arg cfg)/launch/move_group.launch">
    <arg name="allow_trajectory_execution" value="false" />
  </include>
  <node name="joint_states" pkg="joint_state_publisher"
        type="joint_state_publisher" />
  <node name="robot_state" pkg="robot_state_publisher"
        type="robot_state_publisher" />
  <include file="$(arg cfg)/launch/moveit_rviz.launch">
    <arg name="rviz_config"
         value="$(arg cfg)/launch/moveit.rviz" />
  </include>
</launch>
EOF
```

原 demo.launch 含固定本机路径并启动硬件桥接节点。请使用本教程的预览入口；它禁用轨迹执行。下一页为启动与操作方法。

启动与使用

DR-xcobo 2.0 | RViz / OMPL | 中文版

4. 每个新终端：加载环境并启动

```
source /opt/ros/noetic/setup.bash
export ROS_PACKAGE_PATH="$HOME/dr_xcobo_moveit/\
dr-xcobo-2.0-moveit-v1.0:$ROS_PACKAGE_PATH"
roslaunch ~/dr_xcobo_moveit/preview.launch
```

source 加载 ROS；export 告诉 ROS 去哪里找这两个原始包；roslaunch 启动规划和 RViz。反斜杠是续行符，请整段复制。无需另开 roscore。

此预览路径使用已安装的 ROS 节点和原包配置文件，没有自编译节点，因此不需要 catkin_make 或 colcon build；这并不适用于所有 ROS 包。

5. 在 RViz 中规划一条轨迹

① **检查设置**：Global Options → Fixed Frame 选择 base_link；MotionPlanning → Robot Description 使用 robot_description。

② **选择规划组**：Planning Group 改为 arm。原配置中可能显示 left_arm，需要手动切换。若没有规划面板，点击 Add → MotionPlanning。

③ **设置起点和终点**：Start State 选 current；启用 Query Goal State，使用 Interact 拖动目标位姿，或在 Joints 页调整目标关节。起点和终点都应无碰撞。

④ **点击 Plan**：在 Planned Path 中查看轨迹动画。需要逐点查看时，打开 Panels → MotionPlanning - Slider，再点击 Play。

⑤ **退出**：在启动终端按 Ctrl+C。本入口仅规划与预览，不使用 Execute 或 Plan & Execute，也不启动 Gazebo 或真实机器人驱动。

为什么每次都要加载环境?

source 和 export 只影响当前终端。新终端需要重新执行；同一终端无需重复。日常启动不需要再次安装、解压或创建入口文件。

可选：在 ~/.bashrc 末尾添加上方 source 和完整 export 命令（不要加入 roslaunch），再新开终端。只配置一套 ROS 环境；已有设置则不要重复添加。

常见问题

找不到功能包：确认解压目录、大小写及 export 路径，避免另一份同名包被优先找到。
没有模型：确认 meshes 文件夹完整、Fixed Frame 为 base_link。
规划失败：先选择 arm，设置可达且无碰撞的目标，再点击 Plan。

本指南基于原包与官方文档静态核对，尚未在 ROS 环境中运行验证。Ubuntu 20.04 / Noetic 是本包的课程兼容环境；ROS Noetic 和 MoveIt 1 已停止维护。

课程资源 | MoveIt 1 安装 | 官方 RViz 教程